

Mechanical Technology

Welding

Materials

By the end of the topic learners will be able to:-
identify the physical difference between non- ferrous and ferrous metals

Mechanical Technology

Welding

Materials



By the end of the topic learners will be able to:-
identify the physical difference between non- ferrous and ferrous metals

What are materials?

Sheet metal work is a skill and artistry that takes stock sized metals supplied by industry, and it converts it into whatever the mind can conceive

A sheet metalworker will not only know how to use his or her tools, but will also know how materials are handled and what it looks like.

Materials come in shapes and sizes



BAR

A collection of metal bars in various shapes: flat, round, hex, square, and rectangle, arranged diagonally. The bars are rendered in a grayscale style with soft shadows, giving them a three-dimensional appearance. They are positioned against a light gray background, with some bars overlapping others.

Flat
Round
Hex
Square
Rectangle

Round Bar



Round Bar is a long, cylindrical metal bar stock that has many industrial and commercial applications

Standard diameters range from 6mm all the way up to 60 cm

The round bar must be elastic in nature

While choosing a round bar, the most important thing is the end usage



SHEETS

Sheet metal is metal formed by an industrial process into thin, flat pieces

It can be cut and bent into a variety of shapes
hot rolling and cold rolling

Light Sheets

Thin sheets and foils are the last stage in the production of flat products. Thin sheets are between 0.35 and 3.0 mm thick

Heavy Sheets

In case of construction steels, cold rolling only goes up to 3 mm.

From there on, the sheets and plates are hot rolled



Square bars

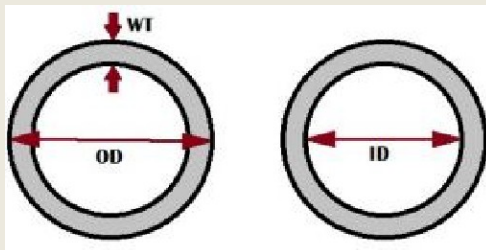


Square steel, squares and square metal bar are a multipurpose steel section mainly used for manufacturing and repairs

Tubing and Pipe

A PIPE is a round tubular to distribute fluids and gases

a TUBE is a round, rectangular, squared, or oval hollow section measured by outside diameter (OD) and wall thickness (WT), expressed in inches or millimetres.



Properties and Uses of Materials

Elasticity

stretching that a metal can do, and then return to its original size

Hardness

power to resist a permanent change in shape

Brittleness

likelihood that a material will fail or fracture under a relatively small shock

Malleability

ability to be hammered, pressed, or rolled into thin sheets without breaking

Ductility

Ductility is the capability of a metal to be permanently bent, twisted, or otherwise manipulated without breaking or cracking

Ferrous Metals

is primarily composed of iron and has magnetic properties

Alloy steel

Carbon steel

Cast iron

Wrought iron

known and used for their strength

a)Mild Steel

b)Galvanised steel

c)Tin plate

d)Zinc aluminium plates

Non-Ferrous Metals

does not have a significant amount of iron in its chemical composition

non-ferrous metals all do have one thing in common: They do not rust

Another characteristic of non-ferrous metals is that they are not magnetic

a)Copper & Copper Plates

. It is a soft, malleable, and ductile metal with very high thermal and electrical conductivity.

b)Brass

is an alloy of copper and zinc, in proportions which can be varied to achieve varying mechanical, electrical, and chemical properties

c)Aluminium

a silvery-white metal, the 13th element in the periodic table. One surprising fact about aluminium is that it's the most widespread metal on Earth

d)Bronze

an alloy consisting primarily of copper, commonly with about 12–12.5% tin and often with the addition of other metals (such as aluminium, manganese, nickel, or zinc) and sometimes non-metals or metalloids such as arsenic, phosphorus or silicon.

Materials Treatment and how it affects the material

Coating is achieved when a thin protective layer is applied to a material

Galvanising

- When steel is brought into contact with molten zinc, a chemical reaction takes place, which bonds zinc on the surface of steel.



- Hot dip galvanization
- Zinc electroplating
- Mechanical plating

Painting



Painting is the practice of applying paint, pigment, colour or other medium to a solid surface.

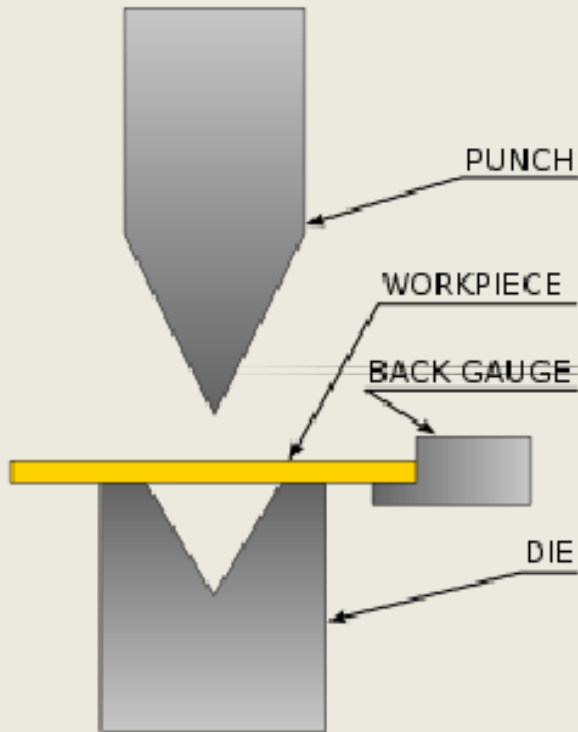
Clean the surface.

Remove loose and peeling paint.

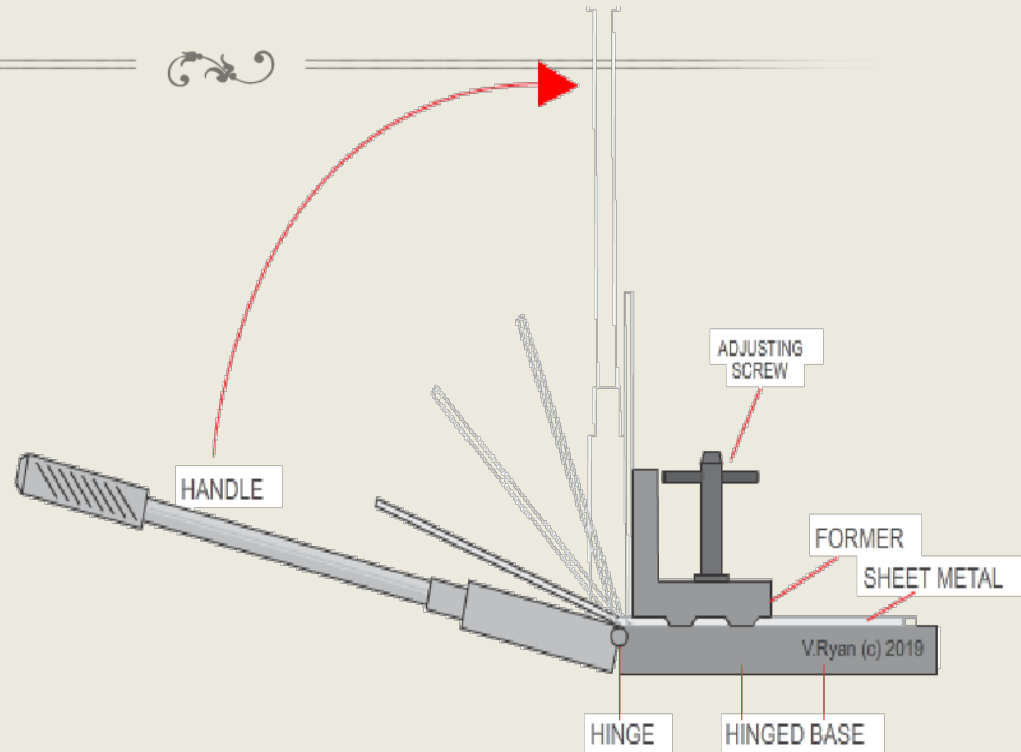
Remove rust.

Repair small holes and dents.

Prime the surface.

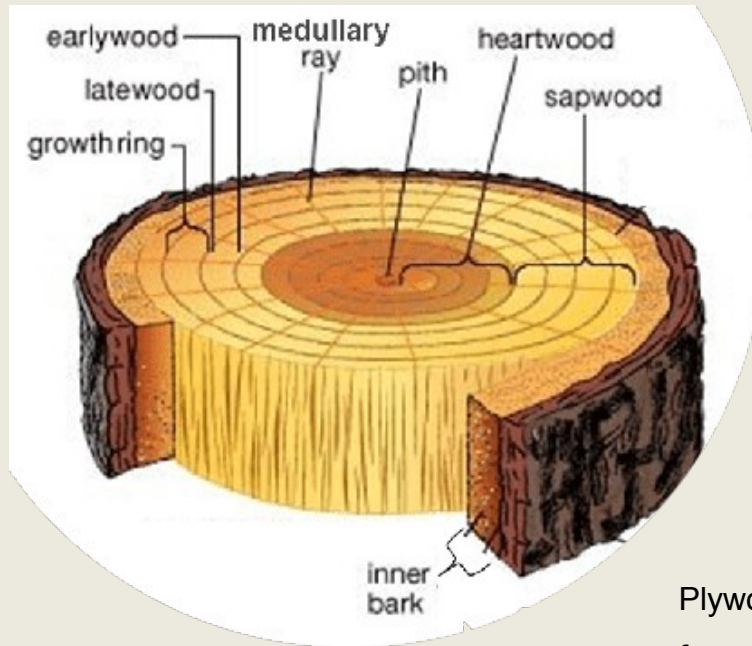


Folding of sheet metal is part of the manufacturing technologies bending and forming. It involves the folding of a surface part of sheet metal.



Plastic change of the shape of sheet metal parts by cold forming along straight lines, simultaneously over the entire length

Other Materials

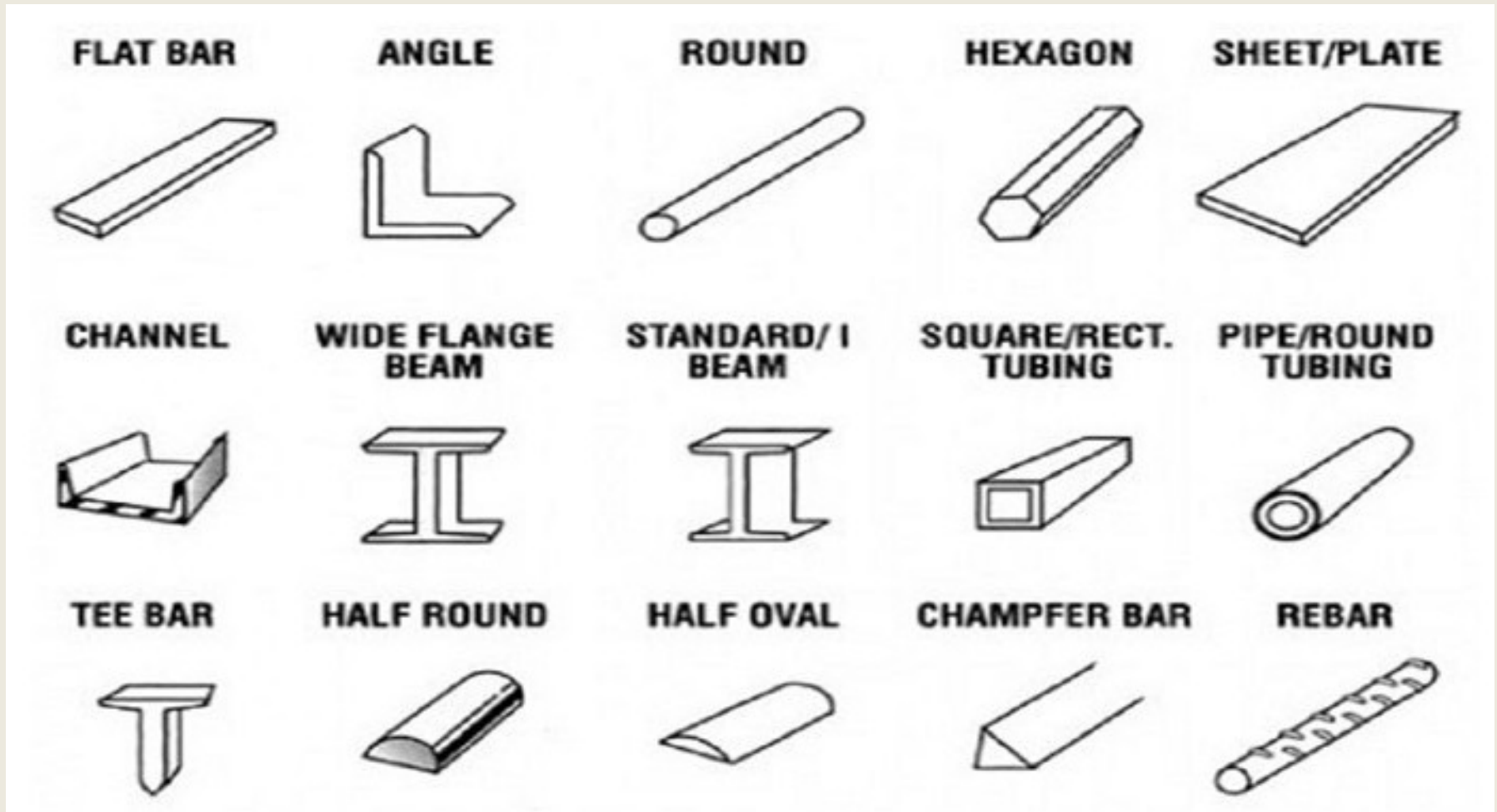


Plywood is a material manufactured from thin layers or "plies" of wood veneer that are glued together with adjacent layers having their wood grain rotated up to 90 degrees to one another. Timber is a type of wood which has been processed into beams and planks.



What Is Mild Steel?

Mild steel is a type of carbon steel with a low amount of carbon, thus the terms low-carbon steel and mild steel



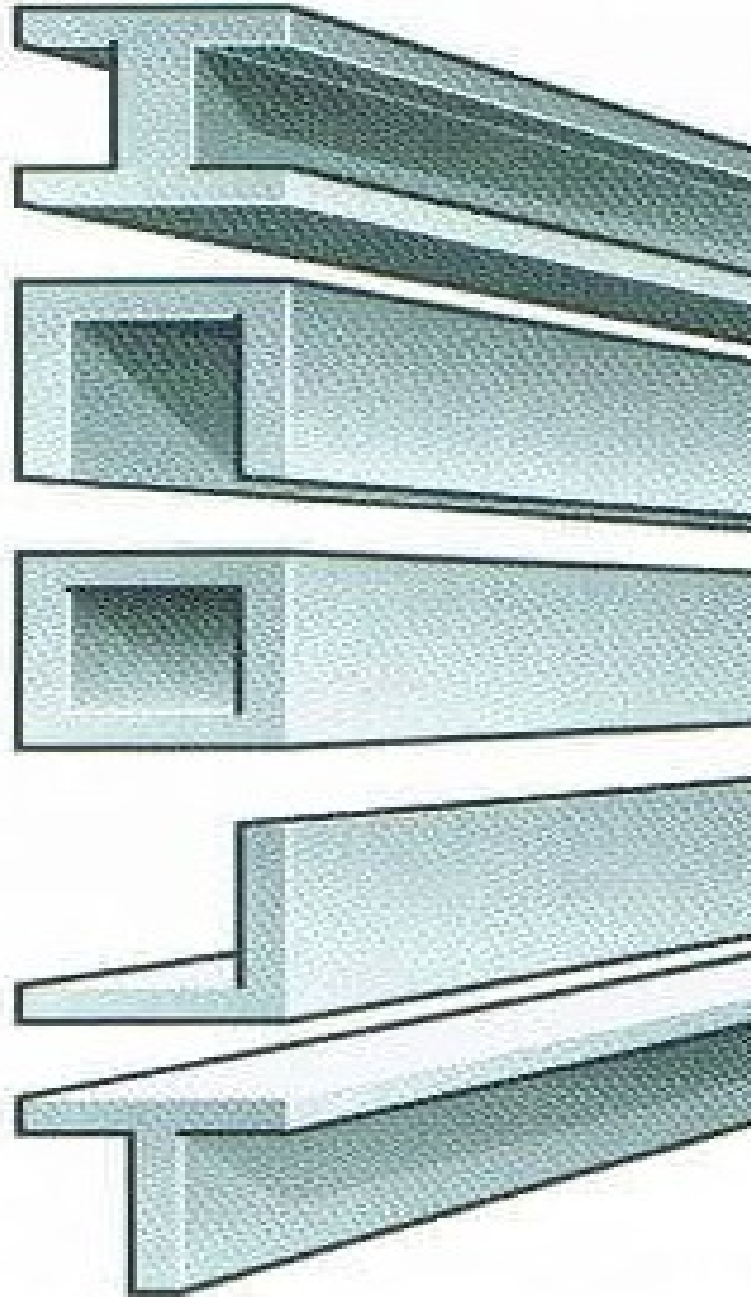
'H'

channel

square

angle

1000 'tee'



VIDEOS FOR MATERIALS



Non Ferrous metals_Types_Properties_Applications_Engineering materials_Aluminium_Zinc_Copper_GTU.mp4

Non-Ferrous Metals List; Its Types, Properties, Uses.

1. Aluminum:

Aluminum is mainly obtained from bauxite ore.

Aluminum is highly resistant to corrosion. When exposed to moist air, aluminum forms a thin film of oxide at the top, which is impervious to air/moisture, and thus saves the metal from further corrosion.

non-ferrous-metals

Properties of Aluminum.

Following are some important properties of this metal.

1. It is silvery-white metal and shows brilliant luster when fresh.

